








Advanced Time Series Analysis and Forecasting

Chapter 15: Review and project defence



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Acronyms used in this material (1/3)

- **ACI**: Adaptive Conformal Inference
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **ALFRED**: ArchivaL Federal Reserve Economic Data
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARDL**: AutoRegressive Distributed Lag (model)
- **ARX**: AutoRegressive model with eXogenous variables
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- **BVAR**: Bayesian Vector AutoRegression
- **CCE**: Common Correlated Effects (estimator)
- **CCM**: Convergent Cross Mapping
- **CEE**: Central and Eastern Europe
- **CPI**: Consumer Price Index
- **CQR**: Conformalized Quantile Regression
- **CRPS**: Continuous Ranked Probability Score
- **CSW**: Chu, Stinchcombe and White (monitoring boundary)
- **CV**: Cross-Validation
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- **DFM**: Dynamic Factor Model
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- **ECB**: European Central Bank
- **EM**: Expectation–Maximisation (algorithm)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EU**: European Union

Acronyms used in this material (2/3)

- **EU-27**: the 27 member states of the European Union
- **EWC**: Equal-Weighted Cosine (estimator)
- **FRED**: Federal Reserve Economic Data
- **FZ0**: Zero-homogeneous Fissler–Ziegel loss (Patton, Ziegel and Chen)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GAS-1F**: One-factor GAS model for VaR and ES
- **GDP**: Gross Domestic Product
- **GMV**: Global Minimum-Variance (portfolio)
- **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR with realised Quarticity
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **HP**: Hodrick–Prescott (filter)
- **IG**: inverse gamma (distribution)
- **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- **ITS**: Interrupted Time Series
- **LLM**: Large Language Model
- **LM**: Lagrange Multiplier
- **LP**: Local Projection
- **LP-IV**: Local Projection with Instrumental Variables
- **LR**: Likelihood Ratio (test)
- **MAE**: Mean Absolute Error
- **MCS**: Model Confidence Set
- **MGARCH**: Multivariate GARCH
- **MIDAS**: MIxed DAta Sampling (regression)
- **ML**: Machine Learning
- **MNZ**: Morley–Nelson–Zivot (2003)
- **MSE**: Mean Squared Error

Acronyms used in this material (3/3)

- **MSFE**: Mean Squared Forecast Error
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **OWA**: Overall Weighted Average (of sMAPE and MASE relative to Naive 2)
- **PCMCI**: PC algorithm + Momentary Conditional Independence
- **PID**: Proportional–Integral–Derivative (controller)
- **PIT**: Probability Integral Transform
- **PMG**: Pooled Mean Group (estimator)
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **QPS**: Quadratic Probability Score
- **RMSE**: Root Mean Squared Error
- **RMSPE**: Root Mean Squared Prediction Error
- **SAV**: Symmetric Absolute Value (CAViaR specification)
- **SC**: Synthetic Control
- **SPA**: Superior Predictive Ability (test)
- **SPF**: Survey of Professional Forecasters
- **STAR**: Smooth Transition AutoRegression
- **SVAR**: Structural Vector AutoRegression
- **TAR**: Threshold AutoRegression
- **TE**: Transfer Entropy
- **TPS**: Taylor, Peel and Sarno (2001)
- **UC-SV**: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- **UC0**: UC model with uncorrelated trend and cycle shocks (Clark 1987)
- **UK**: United Kingdom
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: what does the whole course give you for a research project, and how is the project defended?
  - one chapter that ties Chapters 0–14 and the self-study Chapter 16 to the team project
- **Route**
  - Part I: the course map, one recap per chapter, the overview of methods
  - Part II: the research workflow, replicating a landmark paper, reproducibility, common methodological errors
  - Part III: the oral defence, AI in research, and choosing a project
- There is no written exam: the project and its defence carry 70% of the grade



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Learning outcomes

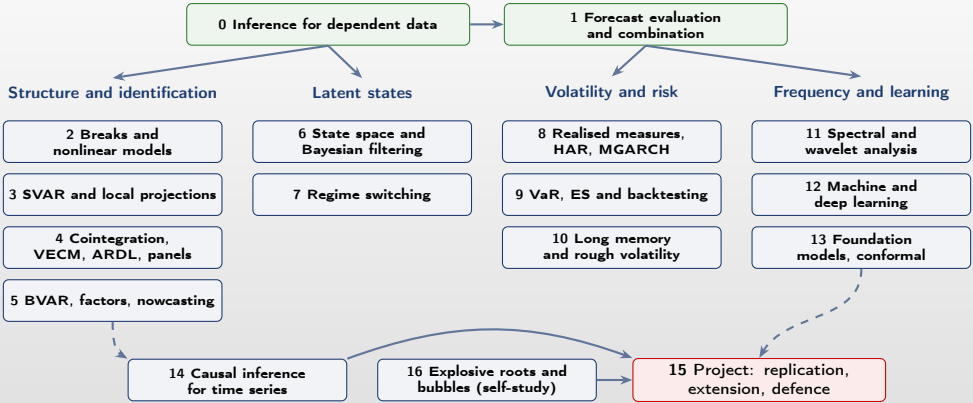
- Place every chapter on the map of methods and state its key idea, key formula and replication result
- Choose the right tool for a research question about dependent data
- Plan a project from the question to the report, with a pre-registration and a power calculation for the evaluation
- Replicate a landmark paper, explain a failed replication and make every number reproducible
- Recognise data snooping, leakage, look-ahead, p-hacking and overclaiming in your own work
- Prepare the individual oral defence and document the use of AI tools

Assessment overview

Component	Weight	Who	Content
Proposal and pre-registration	5%	team	question, paper to replicate, data, evaluation design fixed before estimation
Replication and extension	15%	team	repository, report in the form of a paper, AI_USE.md, AI_ERRORS.md
Oral defence	50%	each student	about 15 minutes: one's own contribution, then questions (format and rubric in Part III)
Chapter quizzes	20%	each student	online, first attempt graded
Attendance	10%	each student	lectures and seminars

- Teams of at most three students; the project can be written and defended in Romanian or in English
- Grades of team members can differ: half of the final grade is individual

The course map



- Chapters 0 and 1 are used everywhere: inference for dependent data and honest out-of-sample evaluation
- Every chapter feeds the project: one landmark paper to replicate and one extension

Five questions for every analysis

- **Target**: which functional is forecast or estimated (mean, quantile, distribution, causal effect)?
  - the loss or score must be consistent for it (Chapters 1 and 9)
- **Dependence**: which standard errors are valid with autocorrelation, heteroskedasticity and persistence?
  - HAC with fixed-b or EWC, block bootstrap, Monte Carlo size at your persistence (Chapter 0)
- **Identification**: what assumption turns a correlation into the claim you make?
  - ordering, sign, instrument, donor pool, regime labels (Chapters 3, 7, 14)
- **Evaluation**: is the design fixed before the results, in real time, with a test that fits?
  - DM–HLN, Clark–West for nested models, MCS and SPA for many
- **Robustness**: does the conclusion survive the forks of the analysis?
  - every chapter ends with a mini-case that varies the specification

Three recurring themes of the course

- **Inference under dependence**
  - long-run variance Ω (Chapter 0, formula on the next slide): the variance of a sum of n dependent terms is about $n\Omega$
  - it reappears in DM tests (Chapter 1), local projections (Chapter 3), cumulative effects (Chapter 14)
  - non-standard limits: sup-tests for breaks, Johansen trace, regime tests, GSADF; critical values from the right distribution
- **Honest out-of-sample evaluation**
  - proper scores and consistent losses, real-time data, blocked validation, evaluation after model release (Chapters 1, 9, 12, 13)
- **Identification and replication**
  - structural shocks, latent states and causal effects are model-dependent; landmark results were replicated on today's data, and some did not survive

Chapter 0: Inference for dependent data (1/2)

- **Key ideas**
  - The precision of a mean or a slope is governed by the long-run variance, not by γ_0
  - HAC with fixed-b or EWC critical values; block bootstraps resample dependence, the wild bootstrap does not
- **Key formula**
  - $\Omega = \gamma_0 + 2\sum_{j\geq 1}\gamma_j = 2\pi f(0)$; $\hat{\Omega}_{NW} = \hat{\Gamma}_0 + \sum_{j=1}^{L}(1 - \frac{j}{L+1})(\hat{\Gamma}_j + \hat{\Gamma}_j')$
  - γ_j: autocovariance at lag j; $f(0)$: spectral density at frequency zero; Ω: long-run variance
  - $\hat{\Gamma}_j$: sample autocovariance matrix at lag j; L: number of lags (bandwidth); $1 - j/(L+1)$: Bartlett weights

Chapter 0: Inference for dependent data (2/2)

- **Replication**
  - Estrella and Hardouvelis (1991): slope 0.52 with classical $t = 4.72$, but fixed-b $t = 2.04$ below its critical value 2.18; 1.12 before 1989, 0.15 afterwards: **partly replicated**
  - White (2000) on 50 moving-average rules for the BET: naive $p = 0.006$, Reality Check $p = 0.06$
- **Common mistakes**
  - i.i.d. standard errors: the naive test rejects 68.1% of true nulls at $\phi = 0.9$, $T = 100$, and still 66.2% at $T = 400$
  - the best of many specifications reported without a search-adjusted p-value

Chapter 1: Forecast evaluation and combination (1/2)

- **Key ideas**
  - Choose the loss or score first: it defines the target (mean, median, quantile, distribution)
  - Calibration with PIT, ranking with proper scores; DM–HLN, Clark–West for nested models, SPA and MCS for many
- **Key formula**
  - $\mathrm{DM} = \bar{d}/\sqrt{\hat{\Omega}_d/P}$, $\mathrm{DM}^* = \mathrm{DM}\sqrt{(P+1-2h+h(h-1)/P)/P}$ against t_{P-1}
  - $\bar{d}$: mean loss differential of the two forecasts; $\hat{\Omega}_d$: its long-run variance; P: number of out-of-sample forecasts; h: horizon
  - DM^*: the HLN small-sample correction, compared with Student-t with $P-1$ degrees of freedom

Chapter 1: Forecast evaluation and combination (2/2)

- **Replication**
  - Meese and Rogoff (1983) on EUR/RON: AR(1) RMSE 1.055 times the random walk, drift 1.006 (Clark–West $p = 0.145$): **replicated**
  - Genre et al. (2013) on the SPF: median 0.999, previous best 1.099 (HLN $p = 0.007$); Atkeson and Ohanian (2001): ratio 1.75 but $p = 0.248$
- **Common mistakes**
  - plain DM with few multi-step forecasts: size 43.6% at $h = 8$, $P = 16$ (HLN: 14.6%)
  - DM on nested models; estimated combination weights for similar forecasts

Chapter 2: Structural breaks and nonlinear models (1/2)

- **Key ideas**
  - A break date chosen from the data is a nuisance parameter: sup, exp and ave tests with their own critical values
  - Breaks and nonlinearity mimic each other: test one allowing for the other
- **Key formula**
  - $\sup_\pi W_T(\pi) \Rightarrow \sup_\pi \|B_p(\pi) - \pi B_p(1)\|^2/[\pi(1-\pi)]$; TAR: $y_t = \phi_{\mathbf{1}}' x_t 1\{q_{t-d} \leq \gamma\} + \phi_{\mathbf{2}}' x_t 1\{q_{t-d} > \gamma\} + \varepsilon_t$
  - $W_T(\pi)$: Wald statistic for a break at the sample fraction π; B_p: a p-dimensional Brownian motion; $\Rightarrow$: convergence in distribution
  - TAR: $\phi_{\mathbf{1}}$, $\phi_{\mathbf{2}}$: coefficients of the two regimes; x_t: lagged regressors; q_{t-d}: threshold variable with delay d; γ: the threshold

Chapter 2: Structural breaks and nonlinear models (2/2)

- **Replication**
  - McConnell and Perez-Quiros (2000): break in 1984Q1, sup-$W = 52.5$; on the sample to 2026 only 10.7; Bai and Perron (2003): 3 breaks at the published dates
  - Hansen (1997): $\hat{\gamma} = 0.261$ (paper 0.302); Taylor, Peel and Sarno (2001): Monte Carlo $p = 0.165$ (paper 0.002): **not replicated**
- **Common mistakes**
  - Chow test at a data-chosen date: size 33–40%; the sup critical value is 8.65, not 3.84
  - in-sample nonlinearity read as a forecasting gain

Chapter 3: Structural VAR and local projections (1/2)

- **Key ideas**
  - A structural VAR is a reduced form plus an identifying assumption; the assumption is the economics
  - Zeros, signs, narratives, heteroskedasticity and instruments buy identification with different assumptions; LP and VAR estimate the same responses
- **Key formula**
  - $u_t = B_{\mathbf{0}}\varepsilon_t$, $\Theta_h = \Phi_h B_{\mathbf{0}}$; proxy: $\mathbb{E}(u_t z_t) = \alpha b_{\mathbf{1}}$; LP: $y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{t+h}$
  - u_t: reduced-form residuals; ε_t: structural shocks; $B_{\mathbf{0}}$: impact matrix; Φ_h: reduced-form responses; Θ_h: structural responses at horizon h
  - proxy: z_t instrument, $b_{\mathbf{1}}$ first column of $B_{\mathbf{0}}$, α relevance; LP: β_h response at h to x_t, w_t controls, ξ_{t+h} error

Chapter 3: Structural VAR and local projections (2/2)

- **Replication**
  - Christiano, Eichenbaum and Evans (1999): industrial production trough -0.27% after 21 months, with the price puzzle; Gertler and Karadi (2015): $F = 17.7$, excess bond premium $+0.14$: **replicated**
  - Kilian (2009): oil-price variance at 60 months: supply 1%, aggregate demand 64%, oil-specific demand 35%
- **Common mistakes**
  - a recursive ordering read as a fact; the wild bootstrap for a proxy SVAR (use the moving-block bootstrap)
  - a short VAR at long horizons: bias 0.71 at $h = 8$ against 0.07 for LP, at a higher variance

Chapter 4: Cointegration: VECM, ARDL and panel data (1/2)

- **Key ideas**
  - Johansen is reduced-rank regression; the rank test is non-standard, depends on the deterministic case and over-rejects in small samples
  - Given the rank, tests on β and α are χ^2; ARDL is a conditional VECM; panels need cross-section dependence first
- **Key formula**
  - $\Delta y_t = \alpha\beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t$, $LR_{tr}(r) = -T \sum_{i>r} \ln(1-\hat{\lambda}_i)$
  - Δ: first difference; β: cointegrating vectors; α: adjustment coefficients; Γ_i: short-run matrices; D_t: deterministic terms with coefficients Φ
  - $LR_{tr}(r)$: trace statistic for rank at most r; $\hat{\lambda}_i$: ordered eigenvalues; T: sample size

Chapter 4: Cointegration: VECM, ARDL and panel data (2/2)

- **Replication**
  - King, Plosser, Stock and Watson (1991): two cointegrating relations, but the great ratios rejected (LR 16.6, $p < 0.001$): **partly replicated**
  - Pesaran, Shin and Smith (1999) on EU-27 consumption: income elasticity 0.69, Hausman $p = 0.300$; the inflation effect is not robust
- **Common mistakes**
  - asymptotic critical values at $T = 50$: size 18.2% against 4.8% for the wild bootstrap
  - the rank read as a measurement: Romanian pass-through, wild bootstrap $p = 0.647$ with four lags

Chapter 5: Bayesian VAR, factor models and nowcasting (1/2)

- **Key ideas**
  - Many series and few observations: shrink (priors) or compress (factors); bigger systems need tighter priors
  - Nowcasting is Kalman filtering with a ragged edge; every revision is a sum of news
- **Key formula**
  - Minnesota: $\text{Var}[(A_l)_{ij}] = \lambda^2/l^2$ if $j = i$, $\vartheta\lambda^2\sigma_i^2/(l^2\sigma_j^2)$ otherwise
  - $(A_l)_{ij}$: effect of variable j at lag l on variable i; the prior variance shrinks with l^2
  - λ: overall tightness; ϑ: extra tightness on other variables; σ_i^2: residual scale of variable i

Chapter 5: Bayesian VAR, factor models and nowcasting (2/2)

- **Replication**
  - Bańbura, Giannone and Reichlin (2010), employment, $h = 1$: 1.06, 0.54, 0.50 for small, medium, large (paper 1.14, 0.54, 0.46): **replicated** for 1971–2003
  - Stock and Watson (2002): industrial production, $h = 12$: 0.58 in 1970–1998, but 1.07 in 1999–2019: the gain vanishes
- **Common mistakes**
  - a homoskedastic BVAR through 2020: relative MSFE 500.06 at 12 months
  - "better nowcasts" from a lower RMSE without a test: Romanian GDP, DM $t = -0.30$, $p = 0.76$

Chapter 6: State space models and Bayesian filtering (1/2)

- **Key ideas**
  - The Kalman filter is Gaussian conditioning applied recursively; smoother, likelihood and simulation smoother follow from it
  - Initialise nonstationary states exactly; stochastic volatility is not GARCH; latent quantities come with model-dependent bands
- **Key formula**
  - $v_t = y_t - Z_t a_t$, $F_t = Z_t P_t Z_t' + H_t$, $K_t = T_t P_t Z_t' F_t^{-1}$, $a_{t+1} = T_t a_t + K_t v_t$
  - a_t, P_t: predicted state and its variance; v_t, F_t: prediction error and its variance; K_t: Kalman gain
  - Z_t: measurement matrix; T_t: transition matrix; H_t: measurement noise variance

Chapter 6: State space models and Bayesian filtering (2/2)

- **Replication**
  - Morley, Nelson and Zivot (2003) on today's GDP vintage: $\hat{\rho} = -0.927$, correlation with the Beveridge–Nelson cycle 0.999; cycle s.d. 2.20 (UC0) against 0.52: **replicated**
  - Stochastic volatility beats GARCH on the S&P 500 by 88.5 log-likelihood points
- **Common mistakes**
  - a zero variance estimate read as a constant level: 64% of estimates are exactly zero when the true variance is zero (pile-up)
  - a "big kappa" initialisation; an output gap without its band

Chapter 7: Regime-switching models (1/2)

- **Key ideas**
  - The Hamilton filter and the Kim smoother are exact recursions for a discrete latent state; EM finds local and degenerate maxima
  - The number of regimes is a non-standard testing problem; regimes, breaks and long memory are observationally close
- **Key formula**
  - $\hat{\xi}_{t|t} = \hat{\xi}_{t|t-\mathbf{1}} \odot \eta_t / 1'(\hat{\xi}_{t|t-\mathbf{1}} \odot \eta_t)$, $\hat{\xi}_{t+\mathbf{1}|t} = \mathrm{P}'\hat{\xi}_{t|t}$, $\mathbb{E}D_i = 1/(1-p_{ii})$
  - $\hat{\xi}_{t|t}$: filtered regime probabilities; η_t: densities of y_t in each regime; $\odot$: elementwise product; 1: vector of ones
  - P: transition matrix; p_{ii}: probability of staying in regime i; $\mathbb{E}D_i$: expected duration of regime i

Chapter 7: Regime-switching models (2/2)

- **Replication**
  - Hamilton (1989) on his data: log-likelihood -181.26 with two implementations, QPS 0.179: **replicated exactly**
  - On today's data the low regime becomes sharp falls: 2008Q4 probability 0.95, 2001 at most 0.01
- **Common mistakes**
  - χ^2 critical values for the number of regimes: LR 71.7, bootstrap 95% quantile 12.32, not 5.99
  - statistical regimes read as economic regimes; smoothed probabilities used to judge real-time dating

Chapter 8: Advanced volatility modelling (1/2)

- **Key ideas**
  - Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
  - Realised measures make volatility observable with a known error; HAR, HARQ and Realized GARCH win one day ahead
- **Key formula**
  - HARQ: $\mathrm{RV}_{t+1} = \beta_0 + (\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t})\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+1}$
  - RV_t: realised variance; $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: its averages over 5 and 22 days; RQ_t: realised quarticity, which measures the error of RV_t
  - $\beta_{dQ} < 0$: the daily weight falls when RV_t is measured with a large error

Chapter 8: Advanced volatility modelling (2/2)

- **Replication**
  - Bollerslev, Patton and Quaedvlieg (2016) on crypto: QLIKE relative to HAR 0.946 (Bitcoin), 0.921 (Ether)
  - Engle, Ghysels and Sohn (2013): industrial production $t = -1.56$; only the realised-variance driver works ($t = 3.86$)
  - Engle, Ledoit and Wolf (2019): GMV volatility 18.33 (DCC-NL) against 18.35 (DCC): the gain does not appear here: **not replicated**
- **Common mistakes**
  - Hessian standard errors with Student-$t(5)$ errors: coverage 73.0% against 91.7% for the sandwich
  - non-robust losses (MAE, MSE on logs) with a noisy volatility proxy

Chapter 9: VaR, ES and backtesting (1/2)

- **Key ideas**
  - A risk forecast is a point forecast of a functional: judge it with a consistent loss; VaR is elicitable, ES only jointly with VaR
  - Backtests are moment tests with estimation risk; comparisons need DM, MCS and Murphy diagrams
- **Key formula**
  - $L_{\mathrm{FZ0}}(y, v, e; \alpha) = -\frac{\mathbf{1}}{\alpha e} 1\{y \le v\}(v - y) + \frac{v}{e} + \ln(-e) - 1$; $\mathrm{VaR}_\alpha = -q_\alpha$
  - y: realised return; v: VaR forecast and e: ES forecast, both as (negative) return quantities; α: the level (e.g. 0.025)
  - L_{FZ0}: a consistent loss for the pair (VaR, ES), lower is better; q_α: the α-quantile of the return

Chapter 9: VaR, ES and backtesting (2/2)

- **Replication**
  - Patton, Ziegel and Chen (2019): GAS-1F loss 0.853 (paper 0.853): the ranking **replicates**; its goodness-of-fit p-values 0.001 and 0.037 (paper 0.242 and 0.313) **do not**
  - Engle and Manganelli (2004) out of sample: only IG passes the DQ test ($p = 0.891$; SAV 0.025)
- **Common mistakes**
  - ranking ES forecasts alone; Normal VaR 1% with a hit rate of 2.2%
  - the $\sqrt{h}$ rule under volatility clustering; "the best ES model" instead of the MCS

Chapter 10: Long memory and rough volatility (1/2)

- **Key ideas**
  - Long memory is a pole at frequency zero; estimate d by (exact) local Whittle and report $\hat{d}(m)$ across bandwidths
  - Volatility is persistent over months and rough over days ($H \approx 0.1$); level shifts mimic long memory
- **Key formula**
  - $f(\lambda) \sim c_f \lambda^{-2d}$, $\gamma(k) \sim c_\gamma k^{2d-1}$; $\mathbb{E}|\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
  - $f(\lambda)$: spectral density at frequency λ; $\gamma(k)$: autocovariance at lag k; d: memory parameter; c_f, c_γ: constants; $\sim$: asymptotic equivalence
  - H: Hurst exponent of log volatility, which measures how rough the path is; Δ: time step; q: moment order; $H < 1/2$: rough paths

Chapter 10: Long memory and rough volatility (2/2)

- **Replication**
  - Gatheral, Jaisson and Rosenbaum (2018) on the S&P 500: $\hat{H} = 0.149$ (0.140 and 0.155 in two halves): **replicated**; Qu (2011): $W = 0.66$, the memory of realised variance survives
  - Rough forecasting gain over HAR at 5 days: QLIKE ratio 0.964, $p = 0.07$: small and horizon-dependent
- **Common mistakes**
  - one bandwidth, one $\hat{d}$; "short memory because $H < 1/2$" (roughness and persistence are different properties)

Chapter 11: Spectral and wavelet analysis (1/2)

- **Key ideas**
  - The spectrum decomposes variance by frequency; filters act on it through their squared gain; multitaper gives low leakage and honest bands
  - Wavelets localise variance and co-movement in time and scale; significance needs Monte Carlo and areawise thinking
- **Key formula**
  - HP cycle gain $G(\omega) = \dfrac{4\lambda(1-\cos\omega)^2}{1+4\lambda(1-\cos\omega)^2}$; coherence $\kappa^2(\omega) = |f_{xy}|^2/(f_x f_y)$
  - ω: frequency; λ: HP smoothing parameter; $G(\omega) \in [0,1]$: share of the variance at ω passed to the cycle
  - f_x, f_y: spectra; f_{xy}: cross-spectrum; $\kappa^2(\omega) \in [0,1]$: squared correlation of x and y at frequency ω

Chapter 11: Spectral and wavelet analysis (2/2)

- **Replication**
  - Cogley and Nason (1995): HP applied to a random walk peaks at 30.1 quarters (7.5 years; simulation 28.6): **replicated**
  - Hamilton (2018) on Romanian GDP: the real-time HP gap correlates 0.57 with the final one and has the wrong sign in 38% of quarters
- **Common mistakes**
  - every wavelet-coherence island read as contagion: BET–DAX 23% significant area against a null 95% quantile of 7.7%
  - pointwise tests across frequencies read as a joint test

Chapter 12: Machine learning and deep learning (1/2)

- **Key ideas**
  - Validation must respect dependence: K-fold only for well-specified autoregressions; otherwise block, purge and keep a final test period
  - Trees and networks rarely beat strong structured baselines alone; global models pool series; seeds and searches are hidden multiple testing
- **Key formula**
  - $\mathrm{OWA} = \frac{1}{2}\left(\mathrm{sMAPE}/\mathrm{sMAPE}_{\mathrm{Naive2}} + \mathrm{MASE}/\mathrm{MASE}_{\mathrm{Naive2}}\right)$; $\mathrm{Att}(Q, K, V) = \mathrm{softmax}(QK'/\sqrt{d_k})V$
  - OWA: the M4 score, relative to the Naive2 benchmark (below 1: better); sMAPE: symmetric mean absolute percentage error
  - Q, K, V: query, key and value matrices; d_k: key dimension; softmax normalises each row to weights that sum to 1

Chapter 12: Machine learning and deep learning (2/2)

- **Replication**
  - Bergmeir, Hyndman and Koo (2018): 5-fold CV error 0.094 against 0.137 out of sample: **replicated**
  - Zeng et al. (2023) and Nie et al. (2023) on Romanian load, $H = 96$: point-token Transformer 0.258, DLinear 0.257, patches 0.226: both papers **replicate**
- **Common mistakes**
  - random K-fold with autocorrelated residuals: validation-to-test error ratio 0.34 (leakage)
  - attention weights read as importance; the best of many seeds reported as the model

Chapter 13: Foundation models and conformal prediction (1/2)

- **Key ideas**
  - Foundation models amortise estimation over a corpus; evaluate after release, pool across series, correct for multiplicity
  - Conformal prediction gives finite-sample marginal coverage under exchangeability; under dependence, ACI restores long-run coverage
- **Key formula**
  - $\hat{q} = S_{(\lceil (n+\mathbf{1})(\mathbf{1}-\alpha) \rceil)}$; ACI: $\alpha_{t+\mathbf{1}} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$
  - $S_{(k)}$: the k-th smallest of n calibration scores; α: the target miss rate; $\hat{q}$: the conformal threshold
  - ACI: α_t working level; γ: step; $\mathrm{err}_t = 1$ after a miss

Chapter 13: Foundation models and conformal prediction (2/2)

- **Replication**
  - Gibbs and Candès (2021) on the S&P 500 (target 0.90): static coverage 0.918, ACI 0.901; Romano, Patterson and Candès (2019): raw quantiles 0.883, CQR 0.915: **replicated**
  - Romanian load: Chronos-2 MAE 0.210 against 0.221 for the expert ARX
- **Common mistakes**
  - cherry-picked countries: Chronos-2 better in 23 of 27 inflation series, significant in 0 after Holm
  - evaluation windows before the model release; conditional coverage claimed instead of tested

Chapter 14: Causal inference for time series (1/2)

- **Key ideas**
  - Granger causality and its relatives (TE, PCMCI, CCM) describe predictive structure; causal claims need assumptions about the assignment
  - Synthetic control uses other units: placebos are the inference, a long pre-period fit is the justification
- **Key formula**
  - $w^* = \arg\min_w (X_{\mathbf{1}} - X_{\mathbf{0}} w)' V (X_{\mathbf{1}} - X_{\mathbf{0}} w)$, $w_j \geq 0$, $\sum_j w_j = 1$; placebo $p = \#\{j : r_j \geq r_{\mathbf{1}}\}/(J+1)$
  - $X_{\mathbf{1}}$, $X_{\mathbf{0}}$: pre-treatment predictors of the treated unit and of the donors; V: predictor weights; w_j: donor weights
  - r_j: post/pre RMSPE ratio of unit j; J: number of donors; $\#\{\cdot\}$: count

Chapter 14: Causal inference for time series (2/2)

- **Replication**
  - Abadie, Diamond and Hainmueller (2015): West German GDP per capita 1588 USD a year (5.6%) below its synthetic control, placebo $p = 0.059$: **replicated**
  - Born et al. (2019) on revised data: gap -1.5% in 2018Q4 (paper -2.4%), UK rank 11 of 24: **not replicated in strength**
- **Common mistakes**
  - Granger causality read as causation; a common driver produces it
  - synthetic control outside the convex hull: classic SC gives 4.18 pp for Romania, the robust estimators 2.7–2.8 pp

Chapter 16: Explosive roots and bubbles (self-study) (1/2)

- **Key ideas**
  - A rational bubble is an explosive component; periodically collapsing bubbles hide from whole-sample tests
  - Critical values must match the design: wild bootstrap under changing volatility, family-wise control for date-stamping
- **Key formula**
  - $\mathrm{BSADF}_{r_2} = \sup_{r_1} \mathrm{ADF}_{r_1}^{r_2}$, $\mathrm{GSADF} = \sup_{r_2} \mathrm{BSADF}_{r_2}$; bubble: $\mathbb{E}_t B_{t+1} = (1+r)B_t$
  - $\mathrm{ADF}_{r_1}^{r_2}$: ADF statistic on the sample fraction $[r_1, r_2]$; BSADF: its sup over the start r_1; GSADF: the sup of BSADF over the end r_2
  - B_t: bubble component of the price; r: interest rate; $\mathbb{E}_t$: expectation given information at t

Chapter 16: Explosive roots and bubbles (self-study) (2/2)

- **Replication**
  - Phillips and Yu (2011): US price-to-rent GSADF 5.75 against the wild critical value 4.47: **replicated**
  - Phillips, Wu and Yu (2011) Nasdaq: 2.92 rejects with Monte Carlo (2.28) but not with the wild bootstrap (3.22); Phillips, Shi and Yu (2015) S&P 500: 2.19 below both: **not replicated**
- **Common mistakes**
  - Monte Carlo critical values after a volatility shift: size 33.8% (wild bootstrap 3.2%)
  - pointwise date-stamping read as a 5% test: false episodes in 32.5% of no-bubble paths

Recap: the sixteen chapters

- Most landmark results replicate in direction on today's data; their strength and their tests often do not
- The same mistakes recur: i.i.d. inference on dependent data, the wrong critical values, a search reported as one test
- Every chapter offers a replication and an extension for the project (▸ Appendix)

Which tool for which question (1/3): inference and evaluation

Research question	Tool	Ch.
Is a mean, a slope or a predictive coefficient different from zero?	HAC with fixed-b or EWC; block bootstrap; Monte Carlo size check	0
Does one of many rules or models beat a benchmark?	Reality Check, SPA, StepM; MCS	0, 1
Is forecast A more accurate than forecast B?	DM–HLN with HAC; Clark–West if nested; Giacomini–White for methods	1
Is a density or interval forecast calibrated?	PIT, Berkowitz; coverage and independence tests; CRPS, log score	1, 13
Is a VaR or (VaR, ES) forecast adequate, and which is best?	Kupiec, Christoffersen, DQ; FZ0 loss, DM, MCS, Murphy diagrams	9
Do my intervals keep their coverage under dependence?	split conformal, CQR, ACI, conformal PID; coverage by regime	13

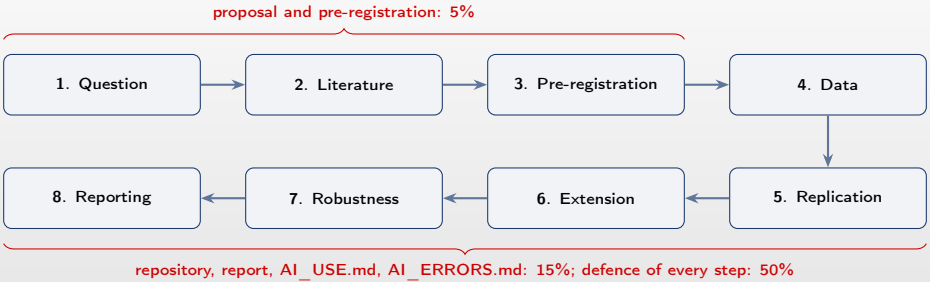
Which tool for which question (2/3): structure and dynamics

Research question	Tool	Ch.
Did the parameters change, and when?	sup/exp/ave tests, Bai–Perron, CSW monitoring, variance-break tests	2
Is the dynamics nonlinear or state-dependent?	TAR, STAR with bootstrap or LM tests; Markov switching with bootstrap LR	2, 7
What is the effect of a structural shock?	SVAR (zeros, signs, proxies), local projections, LP-IV	3
Is there a long-run equilibrium, and who adjusts?	Johansen with bootstrap; tests on β and α; ARDL bounds; panel CCE and PMG	4
How to forecast with many predictors or mixed frequencies?	BVAR with optimised shrinkage, factors, DFM nowcasting, MIDAS	5
What is the unobserved trend, gap or time-varying slope?	state space by ML or Gibbs, simulation smoother, UC-SV, particle filters	6

Which tool for which question (3/3): volatility, frequency, learning, causality

Research question	Tool	Ch.
How to forecast volatility or a covariance matrix?	HAR, HARQ, Realized GARCH; DCC with shrinkage; QLIKE	8
Is the memory long, or is it shifts?	local Whittle $\hat{d}(m)$, Qu test, scaling of log volatility	10
At which frequencies do two series move together?	multitaper coherence and phase, Breitung–Candelon, wavelet coherence	11
Do flexible learners or foundation models beat the baseline?	blocked validation, strong baselines, pooled tests, Holm and BH	12, 13
Does x help predict y, and did a policy have an effect?	conditional Granger, PCMCI; ITS, synthetic control with placebos, CausalImpact	14
Is there a bubble, and when did it start?	GSADF and BSADF with wild bootstrap, family-wise dating	16

The project in eight steps



- The order matters: the evaluation design is fixed before the first estimate, the replication comes before the extension
- Each step leaves a trace in the repository (a file, a commit, a section of the report)

Steps 1–2: the question and the literature

- **A good question** is narrow, falsifiable and answerable with public data
  - "Does the Meese–Rogoff result hold for EUR/RON at horizons of 1–12 months after 2015?" rather than "Can we forecast exchange rates?"
- **The landmark paper**: one of the case studies of Chapters 0–16, or another paper approved in advance
  - identify the table or figure you will reproduce, the sample, the data source and the code, if published
- **The literature**: what was found since, and what is still open
  - every reference checked against its DOI; an AI-suggested reference is a hypothesis until verified Walters and Wilder (2023)
- The extension answers a new question: new data (Romania, the EU, recent years), a new method from the course, or a new test

Step 3: the pre-registration

- A pre-registration fixes the analysis before the data can influence it Nosek et al. (2018)
  - the hypotheses and the primary outcome
  - the data: sources, vintages, sample, transformations, exclusions
  - the evaluation: estimation and test windows, horizons, losses, tests, the benchmark
  - the multiplicity plan: how many comparisons, which correction (Holm (1979), MCS, SPA)
  - the robustness rule: which variations, and what counts as a robust result
- A power calculation for the evaluation: is the test sample long enough to detect the published gain? (mini-case at the end)
- Deviations are allowed but reported: the pre-registration is a commitment to transparency, not a prison
- A template with the seven sections (Appendix)

Steps 4–6: data, replication, extension

- **Data**: official sources, documented once
  - market data from the course repository (EODHD); FRED, Eurostat, ECB, INS, BNR online; record the download date (vintage)
  - real-time questions need real-time data: revised data flatter forecasts (Chapters 1 and 5)
- **Replication**: first the original sample and specification, then today's data
  - a table that puts the published numbers next to yours, with the differences explained
- **Extension**: one new question, answered with the same care
  - new sample, new country, new method of the course, or a stronger test of the original claim

Steps 7–8: robustness and reporting

- **Robustness**: show the distribution of results over reasonable choices, not the best one
  - specification curve or multiverse Simonsohn, Simmons and Nelson (2020), Steegen et al. (2016): windows, lags, donor pools, losses, bandwidths
  - 164 teams testing the same hypotheses on the same data disagree as much as the standard errors suggest Menkveld et al. (2024)
- **Report** in the form of a paper: introduction, data, method, replication, extension, robustness, conclusion
  - every number in the text produced by the code; figures with the course chart rules
  - effect sizes with confidence intervals, not stars; limits stated explicitly
- AI_USE.md and AI_ERRORS.md in the repository (section on AI in research)

Recap: the research workflow

- Question, literature and pre-registration come before any estimate
- Replicate on the original sample first; extend with one new question
- Report the distribution over specifications and every deviation from the plan

Reproduction, replication, robustness

- Clemens (2017) distinguishes
  - **verification**: same data, same code, same numbers (a computational check)
  - **reproduction**: same specification and population, re-coded or with re-collected data
  - **robustness**: a different specification, sample or population: an extension, not a failure if it differs
- In economics, about half of the papers can be reproduced from the authors' files without their help Chang and Li (2022); anomalies in finance often fail on extended samples Hou, Xue and Zhang (2020)
- A spreadsheet error changed a policy debate: Herndon, Ash and Pollin (2014)
- In the course, almost every replication was a reproduction on today's data, followed by a robustness check

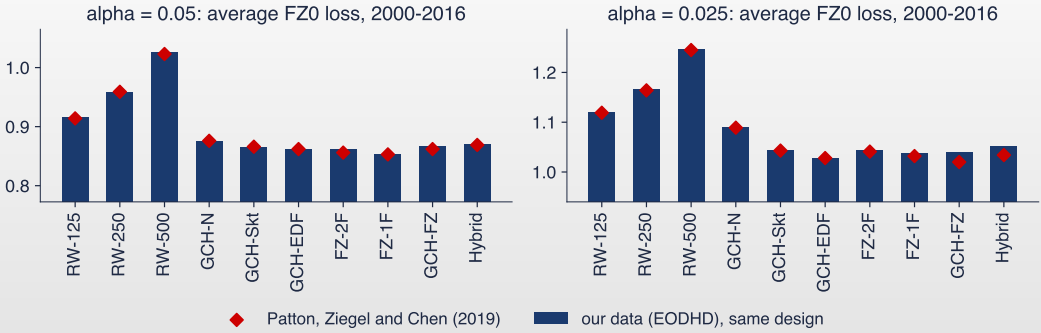
Summary of the course replications (1/2)

Paper	Ch.	Verdict	What decided it
Estrella and Hardouvelis (1991)	0	partly	robust inference and a later sample remove the significance
Meese and Rogoff (1983), Genre et al. (2013)	1	replicated	random walk and the simple average are hard to beat
McConnell and Perez-Quiros (2000), Bai and Perron (2003)	2	replicated	dates reproduced; the 2020 outliers weaken the extension
Hansen (1997)	2	partly	threshold close, delay fragile on revised data
Taylor, Peel and Sarno (2001)	2	not replicated	Monte Carlo p-value; the price index of the paper is not published today
Christiano, Eichenbaum and Evans (1999), Kilian (2009), Gertler and Karadi (2015)	3	replicated	public replication files; same identification
King, Plosser, Stock and Watson (1991)	4	partly	rank reproduced, great ratios depend on the deterministic case
Bańbura, Giannone and Reichlin (2010)	5	replicated	to 2003; fails after 2020 without stochastic volatility
Stock and Watson (2002)	5	partly	gains vanish on 1999–2019
Morley, Nelson and Zivot (2003), Hamilton (1989)	6, 7	replicated	same data; Hamilton's dating does not survive today's sample

Summary of the course replications (2/2)

Paper	Ch.	Verdict	What decided it
Bollerslev, Patton and Quaedvlieg (2016)	8	replicated	in QLIKE; the ranking changes in MSE
Engle, Ghysels and Sohn (2013)	8	partly	the sign survives, the macro effect is not significant
Engle, Ledoit and Wolf (2019)	8	not replicated	few assets relative to the sample: shrinkage has little to do
Patton, Ziegel and Chen (2019)	9	partly	average losses reproduced; tests on a few tail days do not
Gatheral, Jaisson and Rosenbaum (2018), Qu (2011)	10	replicated	scaling stable across halves and markets
Cogley and Nason (1995), Hamilton (2018)	11	replicated	a property of the filter, not of the data
Bergmeir, Hyndman and Koo (2018), Zeng et al. (2023)	12	replicated	protocols reproduced on Romanian load
Gibbs and Candès (2021), Romano, Patterson and Candès (2019)	13	replicated	finite-sample guarantees hold by construction
Abadie, Diamond and Hainmueller (2015)	14	replicated	public data and code; weights to two decimals
Born et al. (2019)	14	not replicated	data revisions, no covariates, permutation inference
Phillips and Yu (2011); Phillips, Shi and Yu (2015)	16	replicated; not replicated	wild bootstrap critical values decide

A successful replication: Patton, Ziegel and Chen (2019)



- Average FZ0 loss of ten (VaR, ES) models, S&P 500, 2000–2016, the design of Patton, Ziegel and Chen (2019): our data (bars) against their Table 8 (diamonds)

ATS_ch9_pzc

Interpreting the two kinds of outcome

- Average losses over 4 000 days are robust to small data differences: GAS-1F 0.853 against 0.853
- Tests that rest on a few dozen tail days are not: goodness-of-fit $p = 0.001$ against 0.242
- The same pattern in Chapter 2 (Taylor, Peel and Sarno (2001): $p = 0.165$ against 0.002) and Chapter 14 (Born et al. (2019): -1.5% against -2.4%)
- Lesson: replicate the estimate and the inference separately; report which of the two survives

Five sources of discrepancy

- **Data vintage**: revised GDP, rebased price indices, series that no longer exist
  - Brexit gap (Chapter 14), the UK CPI of TPS (Chapter 2), the decimals of MNZ (Chapter 6)
- **Sample**: the effect was a feature of its period
  - the term spread (Chapter 0), diffusion indexes (Chapter 5), Hamilton's recessions (Chapter 7)
- **Implementation**: optimiser, starting values, default options, the deterministic case
  - EM maxima (Chapter 7), the great ratios (Chapter 4), diffuse initialisation (Chapter 6)
- **Inference**: the estimate survives, its standard error or critical value does not
  - fixed-b (Chapter 0), goodness-of-fit tests (Chapter 9), wild bootstrap (Chapter 16)
- **Design**: the setting of the paper differs from yours
  - dimension relative to sample for DCC-NL (Chapter 8); the loss function for HARQ (Chapter 8)

How to replicate well

- Obtain the replication package; if none exists, write down every choice the paper leaves open
- Reproduce one number first (a coefficient, a test statistic), then the table, then the figure
- Keep the original sample as a separate run before extending it; change one thing at a time
- Use two implementations when possible (your code and a package), as for Hamilton's model in Chapter 7
- Put published and replicated numbers side by side, with the absolute and relative differences
- A failed replication is a result: report the source of the difference, test the competing explanations, contact the authors politely if needed

The repository

Path	Content
`README.md`	question, data statement, how to run, run time
`requirements.txt`	package versions
`data/raw/`, `data/processed/`	raw files never edited by hand
`code/01_data.py`, `02_replication.py`, ...	numbered scripts, run in order
`notebooks/main.ipynb`	runs in Colab from start to end
`output/tables/`, `output/figures/`	generated by the code, never typed
`report/report.pdf`	the paper
`preregistration.md`, `AI_USE.md`, `AI_ERRORS.md`	the plan and the AI record

- The structure follows the data editors of economics journals Vilhuber (2020); replication packages are now a condition of publication Christensen and Miguel (2018)
- The preregistration.md file is committed before the first estimation commit: the history shows it

Reproducibility checklist

Data and code

- ⊡ every series: source, code, download date
- ⊡ no manual step between raw data and results
- ⊡ one command or one notebook runs everything
- ⊡ random seeds fixed and reported; results stable across seeds
- ⊡ functions tested on a case with a known answer (a simulated process, a published number)

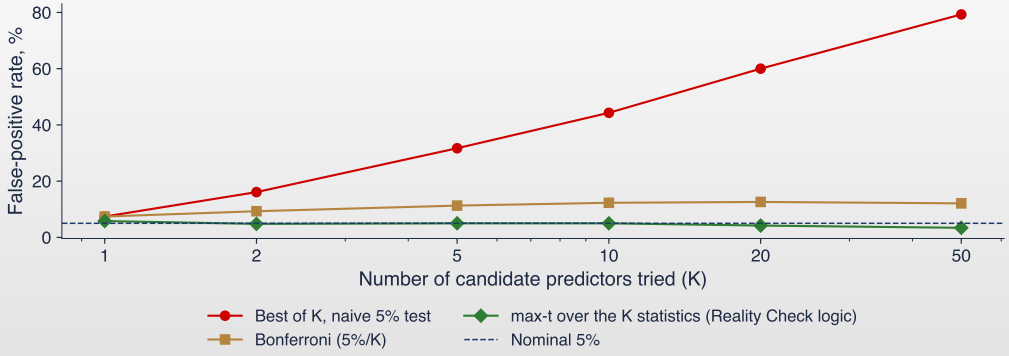
Numbers and report

- ⊡ every number in the report traceable to a file in output/
- ⊡ package versions in requirements.txt
- ⊡ the clean-machine test: a colleague clones and gets the same numbers Peng (2011)
- ⊡ deviations from the pre-registration listed
- ⊡ AI_USE.md and AI_ERRORS.md complete

Data snooping and p-hacking

- **Data snooping**: the same data are used to choose and to test a model; the reported p-value ignores the search White (2000)
  - remedies: Reality Check, SPA Hansen (2005), StepM Romano and Wolf (2005), MCS Hansen, Lunde and Nason (2011), a hold-out period never touched
- **p-hacking**: undisclosed flexibility (samples, controls, transformations, outliers) until $p < 0.05$ Simmons, Nelson and Simonsohn (2011)
  - the "garden of forking paths": no conscious search is needed, data-dependent choices suffice Gelman and Loken (2014)
  - published z-statistics bunch just above 1.96 Brodeur et al. (2016); in finance, a new factor needs $t > 3$ Harvey, Liu and Zhu (2016)
- Persistent predictors add a second problem: HAC tests over-reject, as in Chapter 0 Stambaugh (1999)

A specification search on a series with no predictability



- 1 000 simulated data sets, $T = 250$; K persistent candidate predictors ($\rho = 0.9$, a common factor); HAC t-tests; the share of data sets with at least one "discovery"

ATS_ch15_pitfalls

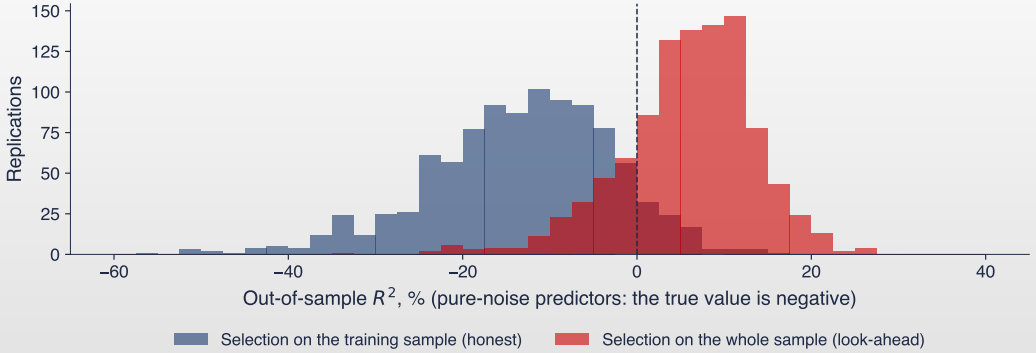
Interpreting the specification search

- Reporting the best of K as if it were the only one: 32% false positives with 5 predictors, 60% with 20, 79% with 50
- Bonferroni does not fix it: 13% at $K = 20$, because each HAC test is already oversized with persistent predictors (7% at $K = 1$)
- The max-t critical value simulated under the null with the same dependence keeps the rate near 5%: 4% at $K = 20$ (the logic of the Reality Check)
- The BET in Chapter 0: naive $p = 0.006$, Reality Check $p = 0.06$
- Pre-register K; report all K results

Leakage and look-ahead

- **Leakage**: information from the evaluation period enters the model Kaufman et al. (2012)
  - selection of predictors, lags or hyperparameters on the whole sample
  - scaling, detrending, filtering (HP, wavelets) or imputing with full-sample statistics
  - random K-fold with dependent residuals or overlapping targets (Chapter 12)
- **Look-ahead**: using data that were not available at the forecast date
  - revised data instead of real-time vintages; publication lags ignored
  - pretrained models whose training data cover the test period (Chapter 13)
- Test: could I have computed this forecast on that date, with that information?

Predictor selection with look-ahead



- 1 000 simulations: $T = 240$, last 60 periods for testing, 100 pure-noise candidate predictors, the 5 most correlated kept, OLS on the training sample

ATS_ch15_pitfalls

Interpreting the leakage experiment

- Honest selection: mean out-of-sample $R^2 = -13.3\%$, positive in 8% of runs: noise does not forecast
- Selection on the whole sample: mean 5.9%, positive in 81% of runs: a "forecasting gain" created by the test period itself
- Out-of-sample evaluation protects you only if every choice is made inside the training window Inoue and Kilian (2005)
- The same mechanism inflated the validation scores of Chapter 12 (ratio 0.34)

Overclaiming

- **Causal language for predictive results**
  - "Granger-causes" is not "causes"; a common driver or timing produces it (Chapter 14)
- **Significance for size**
  - a significant DM test with a 0.3% loss reduction is a small gain; report the effect and its interval
- **Generality for one sample**
  - one market, one period, one specification; say so in the title and the abstract
- **Absence of evidence for evidence of absence**
  - a non-rejection with low power says little: compute the power (Romanian nowcast, $p = 0.76$)



Clive Granger (2008)

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

Recap: common methodological errors

- A search reported as a single test produces discoveries from noise; pre-register K and correct for it
- Every choice inside the training window, every datum available at the forecast date
- Claims no stronger than the design: prediction is not causation, significance is not size

Format of the defence

- Individual, about 15 minutes per student
  - each member is examined separately and receives a separate grade (50% of the final grade)
- 5 minutes: presentation of one's own contribution, the part of the project the student led, on the report and the repository
- 10 minutes: questions on the code, the method and the results
  - the code: open a file, explain a function, say what changes if a choice changes
  - the method: the theory behind it, its assumptions and its alternatives
  - the results: interpret a number, a test, a figure; defend the inference
- Graded with the rubric on the next slide
- Every member must be able to answer questions on the whole project, not only on their part
- No AI tools during the defence; Romanian or English

Grading rubric of the individual defence

Criterion	Points	What earns the points
Mastery of the method	25	states the model and its assumptions, derives the key result, knows when the method fails
Ownership of the code and data	20	explains any function of the repository, the data source and vintage, the effect of changing a setting
Inference and interpretation	25	valid standard errors for dependent data, effect sizes with intervals, correct reading of tests and of non-rejections
Replication and robustness	20	explains differences from the paper, the robustness design and the limits of the conclusion
Research integrity and AI use	10	describes what AI did, which errors were caught and how; deviations from the pre-registration

- The defence grade is the score out of 100 divided by 10; a member who cannot explain the code they submitted cannot pass

What distinguishes the grades

- **9–10**
  - answers “why” and “what if”, not only “what”; anticipates the weak points and has tested them
- **7–8**
  - correct methods and interpretation; hesitates on alternatives or on the asymptotics
- **5–6**
  - can run and describe the code; reads p-values mechanically; limited link to the theory
- **below 5**
  - cannot explain own code or numbers; claims not supported by the results; undeclared AI use

Typical questions with model answers (1/3): the code

- **Question**: Where in your code is the information set of the forecast for date t fixed?
  - **Model answer**: In the walk-forward loop: the estimation window ends at $t - h$, and scaling, selection and tuning are recomputed inside the loop (shows the lines)
- **Question**: How many HAC lags do you use, and why?
  - **Model answer**: The Newey–West rule $\lfloor 4(T/100)^{2/9} \rfloor$ (T: sample size; $\lfloor \cdot \rfloor$: integer part), at least $h - 1$ for overlapping h-step errors; checked with fixed-b critical values (Chapter 0)
- **Question**: What happens if you change the seed?
  - **Model answer**: We reran with ten seeds: the ranking is unchanged, the loss changes by less than its standard error (a table in the report)

Typical questions with model answers (2/3): the results

- **Question**: Your DM test gives $p = 0.03$: is your model better?
  - **Model answer**: Better in the pre-registered loss over this sample: the mean loss differential is X with a HAC interval; the models are not nested, so DM is valid; with several models we report the MCS
- **Question**: Your replication differs from Table 2 of the paper. Why?
  - **Model answer**: On the original sample we match to two decimals; the difference appears with the current data vintage, so it is revision, not code (shows the two runs)
- **Question**: What result would have made you reject your hypothesis?
  - **Model answer**: The pre-registration states it: a placebo date with a similar effect, or an effect that changes sign across the donor pools

Typical questions with model answers (3/3): the chapters

- **Question**: Why can you not use χ^2 critical values to test two regimes against one?
  - **Model answer**: Under the null the transition probabilities are not identified, a parameter is on the boundary and the score is zero: the bootstrap or Hansen's and Garcia's methods are needed (Chapter 7)
- **Question**: Why is K-fold cross-validation acceptable for your AR model but not for your random forest?
  - **Model answer**: For a well-specified autoregression the residuals are uncorrelated, so the folds are valid Bergmeir, Hyndman and Koo (2018); the forest's residuals are autocorrelated, so we block and purge (Chapter 12)
- **Question**: Is a significant Granger test a causal effect?
  - **Model answer**: No: it is predictive relative to an information set; a causal claim needs an identifying assumption about the assignment, such as an instrument or a synthetic control with placebos (Chapter 14)

Common weaknesses of project defences

- Code written by someone else (a colleague or an AI assistant) that the presenter cannot explain
- A p-value read as the probability that the hypothesis is true
- A forecasting gain without a test, or a test without the size of the gain
- Not knowing the data: source, frequency, vintage, transformations, sample dates
- Numbers in the report that the repository does not reproduce
- Defending every result instead of explaining its limits

AI_USE.md: what was asked and what was kept

- AI tools are allowed and declared; undeclared use counts as plagiarism
- For each use, one entry
  - the tool and version, the date, the step of the workflow
  - the prompt (or a faithful summary of it)
  - what was kept, changed or rejected, and how it was checked
- Useful uses
  - literature search with verified DOIs; first drafts of code with tests; a hostile-referee critique of the design Korinek (2023)
- Risky uses
  - references, numbers and dates taken without checking Walters and Wilder (2023); text that claims more than the results
  - the illusion of understanding: fluent output mistaken for knowledge Messeri and Crockett (2024)

AI_ERRORS.md: at least three errors caught

Type	Example from the course	How it is caught
invented reference	a plausible title with a DOI that resolves to another paper	Crossref lookup: title and authors must match
wrong formula	ES scored alone with a quantile loss (Chapter 9)	derivation on paper; a simulation with known answer
leakage in code	a scaler fitted on the whole sample before the split	the date test: recompute a forecast with data up to its date only
wrong default	a χ^2 critical value for a sup-test (Chapter 2)	read the documentation and the paper; reproduce a published value
overclaiming	"Granger causality proves that ..." (Chapter 14)	compare each sentence with the design

- Each entry: what the tool produced, why it was wrong, how it was detected, what replaced it

A verification protocol

- References: every DOI resolved and the title matched; the claimed result found in the paper
- Code: run on a simulated process with known parameters before the real data; compare with a second implementation
- Numbers: every number in the report regenerated by the repository; none copied from a chat
- Dates and facts: from official sources (Eurostat, BNR, Monitorul Oficial), not from the model's memory
- Text: each claim checked against the design (prediction, association or causation)
- At the defence you explain everything yourself: what you cannot explain should not be in the project

The open question

- **Which landmark results in time series econometrics survive on new data, and why do some not?**
  - the course replications point to vintages, samples and inference; a systematic answer across papers does not exist
- AI tools change the cost of each step
  - literature maps and code drafts in hours Wang et al. (2023); fully automated pipelines have been proposed Lu et al. (2024)
  - the scarce resource becomes judgement: the design, the checks, the interpretation
- Your project is one data point in this question: a replication with a clear verdict and a documented reason

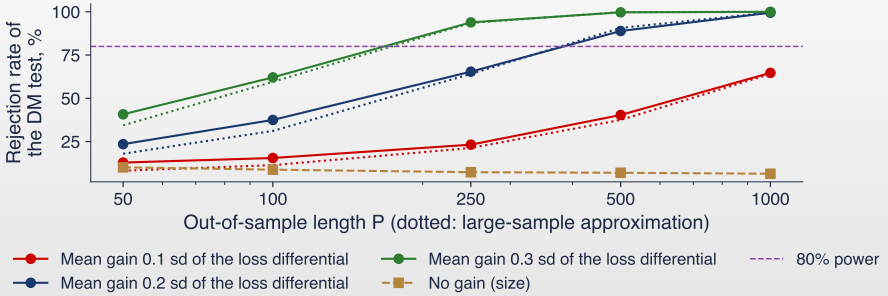
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List replications of Meese and Rogoff (1983) for Central and Eastern European currencies since 2010, with DOIs, horizons and verdicts.` Then check every DOI
  - **design**: `Given a loss differential with autocorrelation 0.3, how many out-of-sample months are needed for 80% power to detect a mean gain of 0.2 standard deviations?`
  - **code and replication**: ask for a DM–HLN function, then test it on simulated data with a known answer
  - **critique**: `Act as a hostile referee of this pre-registration: where can the results leak into the design?`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- The paper exists, says what is claimed, and its data are available today
- The power calculation uses the dependence of your own loss differential, estimated on the training period
- The formula behind an AI answer: a power claim without the long-run variance is wrong for dependent data
- The extension is feasible in one semester with the course tools
- The verdict criteria (replicated, partly, not) are written before the replication is run

Mini-case: power of the Diebold–Mariano test and the evaluation sample size



- Rejection rate of the DM–HLN test (5%, two-sided, 2 000 simulations) when the loss differential is a mean gain δ plus an AR(1) noise with coefficient $\rho = 0.3$ and unit variance
- Dotted: the approximation $\Phi\left(\delta\sqrt{P/\Omega} - 1.96\right)$; Φ: standard Normal distribution function; P: number of out-of-sample observations; Ω: long-run variance of the differential

ATS_ch15_power

Interpreting the mini-case

- A gain of 0.2 standard deviations is detected with probability 38% at $P = 100$ and 89% at $P = 500$; 80% power needs about 364 observations
- A gain of 0.1 needs about 1458: more than 120 years of monthly data; a gain of 0.3 about 162
- The sample-size rule $P \approx \Omega\,(z_{0.975} + z_{0.8})^2/\delta^2$, with $\Omega = (1+\rho)/(1-\rho) = 1.86$: dependence multiplies the sample you need
  - $z_{0.975} = 1.96$, $z_{0.8} = 0.84$: Normal quantiles for a 5% two-sided test and 80% power; δ: the mean gain in standard deviations
- With few observations the test is also oversized: 10.1% at $P = 50$
- Choose a question your data can answer: monthly macro data rarely detect small gains, daily or intraday data can

Project idea

- **Choosing the project: four criteria**
  - a landmark paper with public data or a precise data description, and a number to match
  - an extension that answers one new question, preferably on Romanian or EU data
  - enough power for the evaluation you plan (the mini-case)
  - a verdict rule written before the first estimate
- **Example**: why do some replications fail? A replication audit of three course papers
  - replicate Taylor, Peel and Sarno (2001), Patton, Ziegel and Chen (2019) and Born et al. (2019) on the original and on today's data; attribute each difference to vintage, sample, implementation or inference
  - extend: archived vintages (ALFRED, OECD Economic Outlook), a specification curve, a power analysis of each original test
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Five questions for every analysis: target, dependence, identification, evaluation, robustness
- Most landmark results replicate in direction; their strength and their tests depend on vintage, sample and inference
- Pre-register the design, compute the power, and make every number reproducible from one command
- Searches, leakage and look-ahead create discoveries from noise; causal words need causal designs
- The defence is individual: explain the method, the code, the inference and the limits yourself

Self-assessment

Questions

- Which assumption identifies the main result of your project?
- How many specifications did you try, and how does your inference account for them?
- Could each forecast have been computed on its date?
- What is the power of your main test?
- Which of your numbers would change with a new data vintage?

Further study

- Christensen and Miguel (2018); Clemens (2017); Menkveld et al. (2024)
- Diebold (2015); Lazarus et al. (2018)
- Self-study Chapter 16 and its quiz

Appendix: project seeds by chapter (1/2)

Ch.	Replicate	Extend
0	Estrella and Hardouvelis (1991)	the euro area and Romania; fixed-b inference; stability across decades
1	Meese and Rogoff (1983)	CEE currencies, density forecasts, real-time macro fundamentals
2	Bai and Perron (2003)	breaks in Romanian inflation and the policy rate; real-time monitoring
3	Gertler and Karadi (2015)	ECB surprises and Romanian financial conditions with LP-IV
4	Pesaran, Shin and Smith (1999)	interest-rate pass-through in the EU with CCE
5	Bańbura, Giannone and Reichlin (2010)	a large BVAR with stochastic volatility for Romania, nowcasting with news
6	Morley, Nelson and Zivot (2003)	the Romanian output gap and its real-time revisions
7	Hamilton (1989)	real-time recession probabilities for the euro area
8	Bollerslev, Patton and Quaedvlieg (2016)	HARQ for the BET and EUR/RON with realised measures

Back

Appendix: project seeds by chapter (2/2)

Ch.	Replicate	Extend
9	Patton, Ziegel and Chen (2019)	CEE indices; the power of the backtests
10	Gatheral, Jaisson and Rosenbaum (2018)	roughness of CEE and crypto volatility; robustness to noise
11	Cogley and Nason (1995), Hamilton (2018)	real-time output gaps of the EU members
12	Zeng et al. (2023)	EU electricity load; global models; benchmark comparisons
13	Gibbs and Candès (2021)	conformal VaR for the BET; coverage by regime
14	Abadie, Diamond and Hainmueller (2015), Born et al. (2019)	tax and energy measures in CEE with synthetic controls
16	Phillips and Yu (2011)	house prices and rents in EU capitals
15	Taylor, Peel and Sarno (2001), Patton, Ziegel and Chen (2019), Born et al. (2019)	a replication audit (the project idea of this chapter)

- Part C of every seminar contains a further project seed with its data

Back

Appendix: a pre-registration template

- **1. Question and hypotheses**: the primary hypothesis, in one sentence; secondary hypotheses
- **2. Paper to replicate**: the table or figure, the sample, what counts as "replicated", "partly", "not replicated"
- **3. Data**: series, sources, vintage date, frequency, transformations, exclusions
- **4. Models**: benchmark and competitors, estimation method, how hyperparameters are chosen (inside the training window)
- **5. Evaluation**: windows (rolling or expanding), horizons, losses, tests, multiplicity correction, power calculation
- **6. Robustness**: the list of variations and the rule for a robust result
- **7. Team**: who leads which part (the basis of the individual defence)

Back

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